

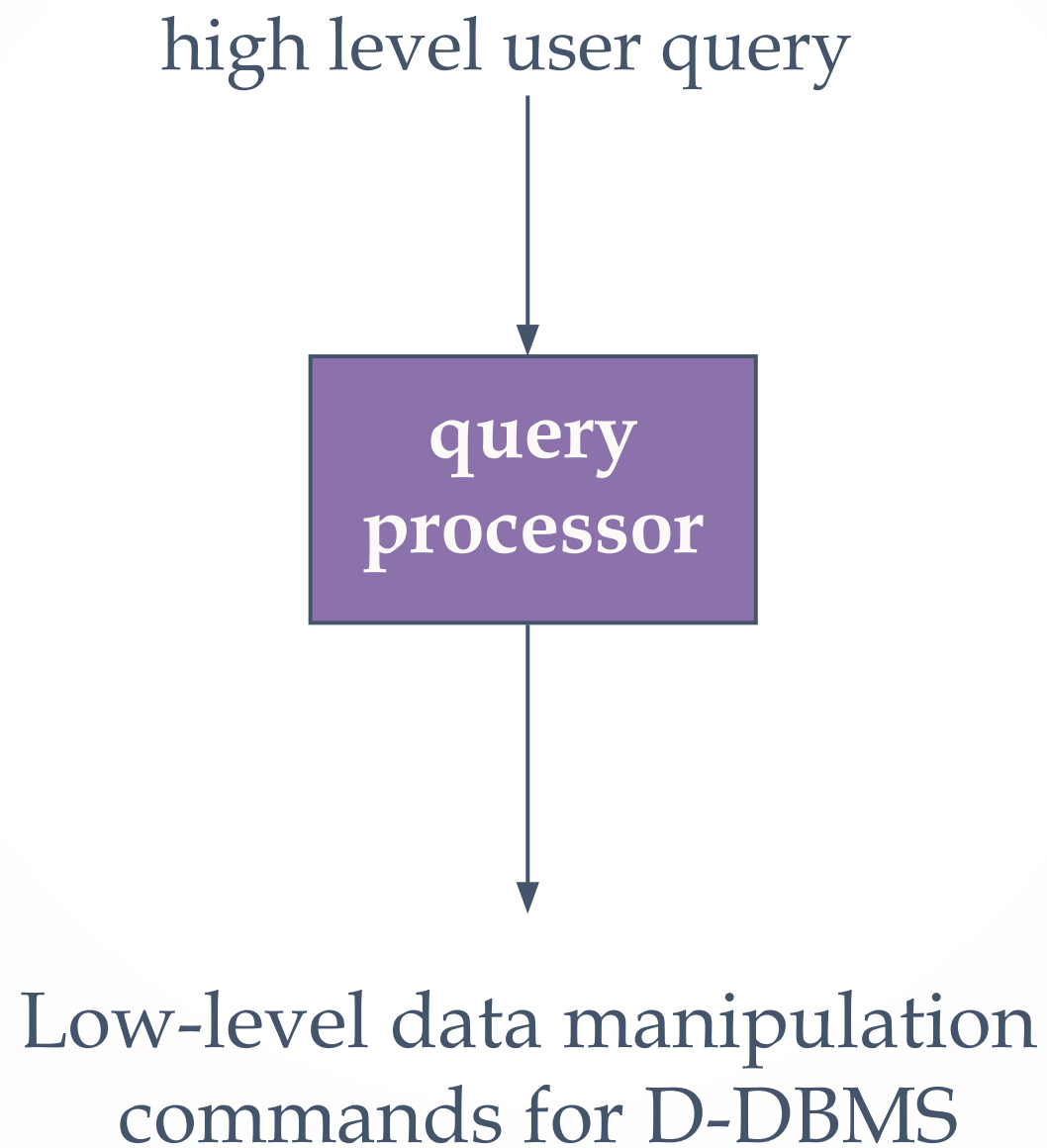
Query Processing in D-DBMS

Lecturer: Nguyen Dinh Thanh

Email: dinhthanh@hcmut.edu.vn

More info: <https://nguyendinhthanh.net>

Query Processing in a DDBMS



Query Processing Components

- Query language that is used
 - SQL: “intergalactic dataspeak”
- Query execution methodology
 - The steps that one goes through in executing high-level (declarative) user queries.
- Query optimization
 - How do we determine the “best” execution plan?
- We assume a homogeneous D-DBMS

Selecting Alternatives

<i>EMP</i> (<i>ENO</i> , <i>ENAME</i> , <i>TITLE</i>) <i>ASG</i> (<i>ENO</i> , <i>PNO</i> , <i>RESP</i> , <i>DUR</i>)
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"Find the names of employees who are managing a project"

```
SELECT ENAME
FROM    EMP, ASG
WHERE    EMP.ENO = ASG.ENO
AND      RESP = "Manager"
```

Strategy 1

$$\Pi_{ENAME}(\sigma_{RESP="Manager" \wedge EMP.ENO=ASG.ENO}(EMP \times ASG))$$

Strategy 2

$$\Pi_{ENAME}(EMP \bowtie_{ENO} (\sigma_{RESP="Manager"}(ASG)))$$

Strategy 2 avoids Cartesian product, so may be "better"

What is the Problem?

$$\Pi_{\text{ENAME}}(\text{EMP} \bowtie_{\text{ENO}} (\sigma_{\text{RESP}=\text{"Manager"}}(\text{ASG})))$$

Site 1

Site 2

Site 3

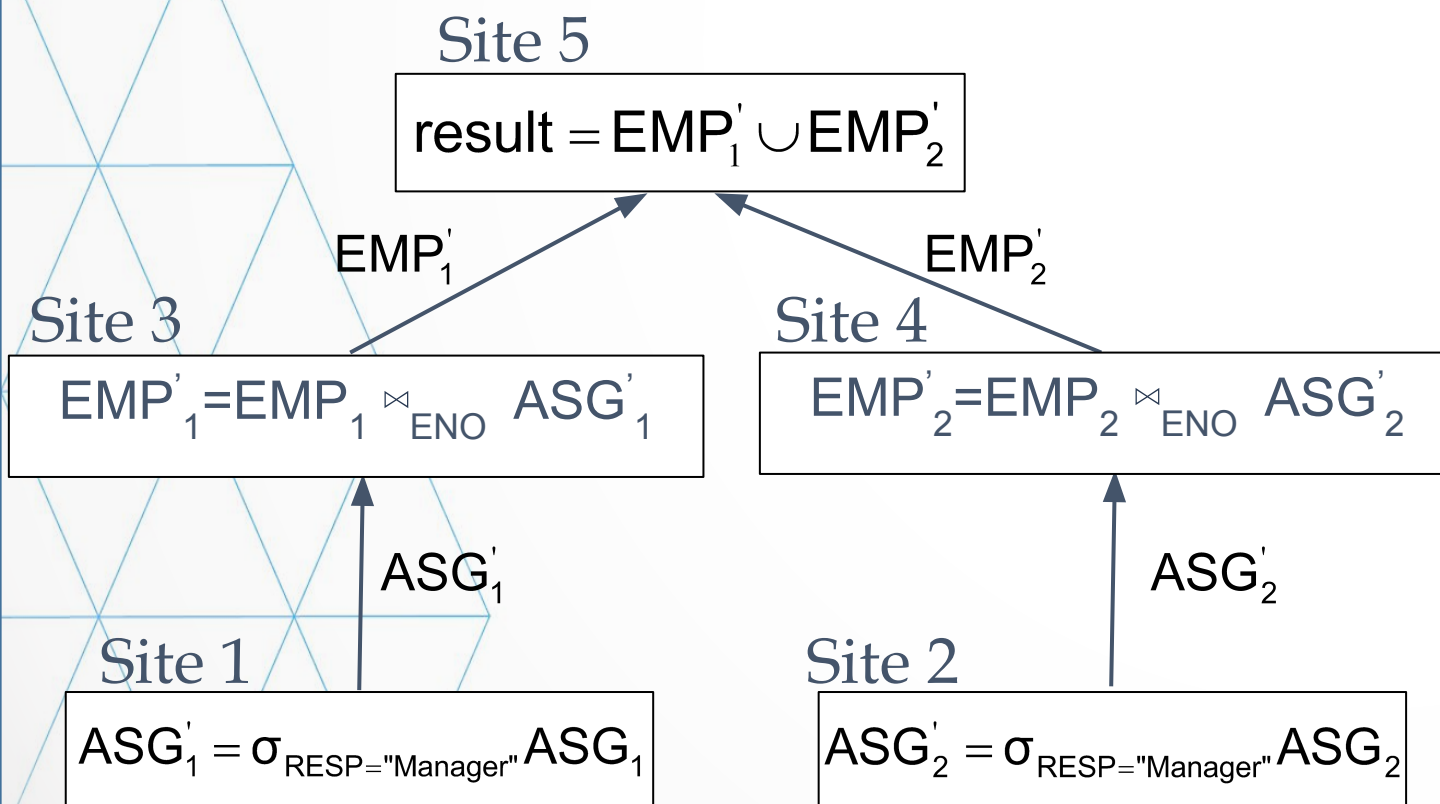
Site 4

Site 5

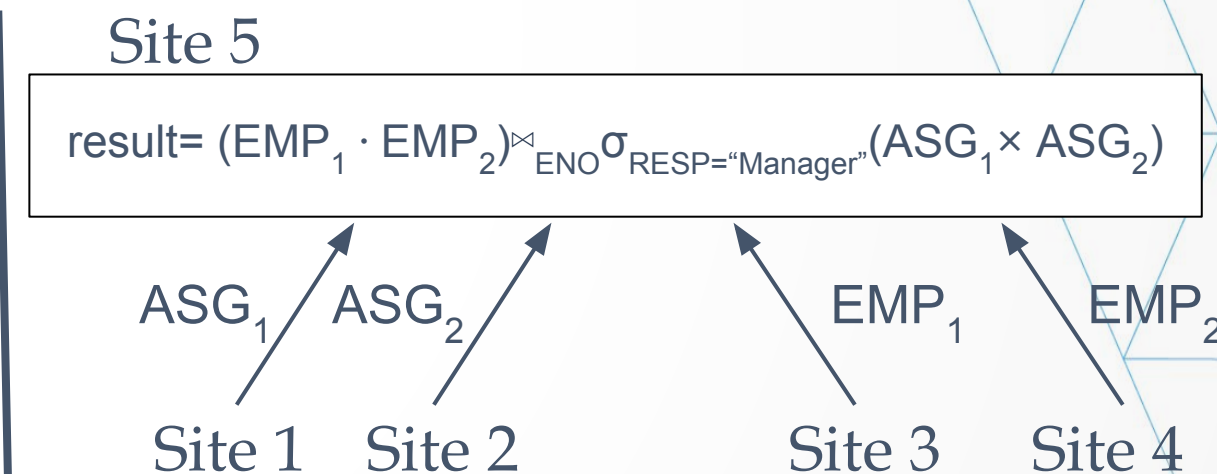
Result

$$\text{ASG}_1 = \sigma_{\text{ENO} \leq \text{"E3"}}(\text{ASG}) \quad \text{ASG}_2 = \sigma_{\text{ENO} > \text{"E3"}}(\text{ASG}) \quad \text{EMP}_1 = \sigma_{\text{ENO} \leq \text{"E3"}}(\text{EMP}) \quad \text{EMP}_2 = \sigma_{\text{ENO} > \text{"E3"}}(\text{EMP})$$

Execution Strategy 1



Execution Strategy 2



Which is better? Strategy 1 or Strategy 2

Cost of Alternatives

- Assume
 - $size(EMP) = 400, size(ASG) = 1000$
 - tuple access cost = 1 unit; tuple transfer cost = 10 units
 - 20 managers in ASG relation
 - Data is uniformly distributed among sites
- Strategy 1
 - produce ASG': $(10+10) * \text{tuple access cost}$ **20**
 - transfer ASG' to the sites of EMP: $(10+10) * \text{tuple transfer cost}$ **200**
 - produce EMP': $(10+10) * \text{tuple access cost} * 2$ **40**
 - transfer EMP' to result site: $(10+10) * \text{tuple transfer cost}$ **200**
 - Total Cost 460**
- Strategy 2
 - transfer EMP to site 5: $400 * \text{tuple transfer cost}$ 4,000
 - transfer ASG to site 5: $1000 * \text{tuple transfer cost}$ 10,000
 - produce ASG': $1000 * \text{tuple access cost}$ 1,000
 - join EMP and ASG': $400 * 20 * \text{tuple access cost}$ 8,000
 - Total Cost 23,000**

Query Optimization Objectives

- Minimize a cost function

I/O cost + CPU cost + communication cost

These might have different weights in different distributed environments

- Wide area networks
 - communication cost may dominate or vary much
 - bandwidth
 - speed
 - high protocol overhead
- Local area networks
 - communication cost not that dominant
 - total cost function should be considered
- Can also maximize throughput

Complexity of Relational Operations

- Assume
 - relations of cardinality n
 - sequential scan

Operation	Complexity
Select Project (without duplicate elimination)	$O(n)$
Project (with duplicate elimination) Group	$O(n * \log n)$
Join Semi-join Division Set Operators	$O(n * \log n)$
Cartesian Product	$O(n^2)$

Query Optimization Issues – Types Of Optimizers

- Exhaustive search

- Cost-based
- Optimal
- Combinatorial complexity in the number of relations

- Heuristics

- Not optimal
- Regroup common sub-expressions
- Perform selection, projection first
- Replace a join by a series of semijoins
- Reorder operations to reduce intermediate relation size
- Optimize individual operations

Query Optimization Issues – Optimization Granularity

- Single query at a time
 - Cannot use common intermediate results
- Multiple queries at a time
 - Efficient if many similar queries
 - Decision space is much larger

Query Optimization Issues – Optimization Timing

- Static
 - Compilation □ optimize prior to the execution
 - Difficult to estimate the size of the intermediate results ⇒ error propagation
 - Can amortize over many executions
 - R*
- Dynamic
 - Run time optimization
 - Exact information on the intermediate relation sizes
 - Have to reoptimize for multiple executions
 - Distributed INGRES
- Hybrid
 - Compile using a static algorithm
 - If the error in estimate sizes > threshold, reoptimize at run time
 - Mermaid

Query Optimization Issues – Statistics

- Relation
 - Cardinality
 - Size of a tuple
 - Fraction of tuples participating in a join with another relation
- Attribute
 - Cardinality of domain
 - Actual number of distinct values
- Common assumptions
 - **Independence** between different attribute values
 - **Uniform distribution** of attribute values within their domain

Query Optimization Issues – Decision Sites

- Centralized
 - Single site determines the “best” schedule
 - Simple
 - Need knowledge about the entire distributed database
- Distributed
 - Cooperation among sites to determine the schedule
 - Need only local information
 - Cost of cooperation
- Hybrid
 - One site determines the global schedule
 - Each site optimizes the local subqueries

Query Optimization Issues – Network Topology

- **Wide area networks (WAN)** – point-to-point
 - Characteristics
 - Low bandwidth
 - Low speed
 - High protocol overhead
 - Communication cost will dominate; ignore all other cost factors
 - Global schedule to minimize communication cost
 - Local schedules according to centralized query optimization
- **Local area networks (LAN)**
 - Communication cost not that dominant
 - Total cost function should be considered
 - Broadcasting can be exploited (joins)
 - Special algorithms exist for star networks

Distributed Query Processing Methodology

